CPE 381 Classwork 02: Mathematical Preliminaries

Student Name (as in Canvas): _____

A Number: _____

Points: 20 Duration: 15 minutes Due: August 26, 2026 (In-Class)

1. Consider the complex number z on Figure 1. What is the complex conjugate $\overline{z}$? **(3 Points)**

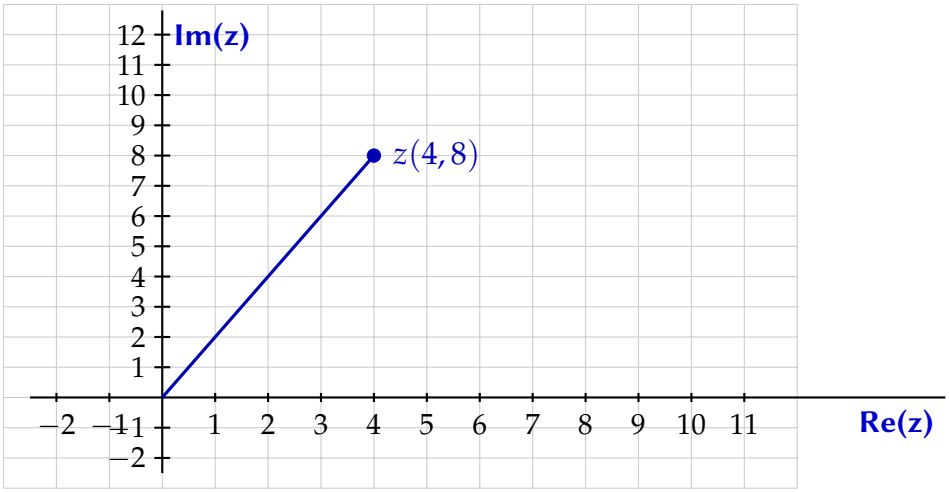


Figure 1: Complex z.

Answer

2. If $z = \sqrt{3} + i$, what is z^4 in polar form? Use De Moivre's Theorem. **(6 Points)**

Answer

3. What is the numerical value of $2(\sin^2\theta + \cos^2\theta)$? **(3 Points)**

Answer

4. If $\sin(\theta) = \frac{12}{13}$, and θ lies in the second quadrant, find the value of $\sec(\theta) + \tan(\theta)$. **(8 Points)**

Answer